FArms: Federated Multi-Modal Learning with Simulating Arms for Missing Modalities

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Abstract

We introduce a multi-modal learning framework that handles arbitrary missing modalities **w**ith learnable simulating **arms** (wArms), which supports robust training, validation, and inference on incomplete data by reconstructing missing modality embeddings from available ones, leveraging frozen pre-trained encoders, pairwise and fusion-based simulators, and a two-stage optimization strategy. The approach is promoted to federated setting (FArms), and performs effectively across both classification and regression tasks. Extensive experiments on MM-IMDb and CMU-MOSI under severe and complex missing modality configurations demonstrate that FArms consistently achieves superior and stable performance on both datasets. While some baselines perform reasonably well on one dataset or specific settings, they suffer noticeable degradation under heavy missing regimes. In the federated setting, FArms converges effectively with only 10 communication rounds while maintaining competitive communication overhead. Code is available at: https://github.com/for-reviewers-arz/FArms_wArms/tree/main.

1. Introduction

Multimodal learning has become essential for leveraging complementary information across heterogeneous data sources, enabling robust performance in diverse tasks such as classification and regression. However, real-world datasets frequently suffer from missing modalities due to sensor failures, privacy constraints, or data collection limitations, leading to significant performance degradation when models assume complete modality availability during training or inference (Ma et al., 2022; 2021; Dai et al., 2025). Existing centralized approaches have made progress in handling missing modalities through generation-based reconstruction (Liu et al., 2024; Zhao et al., 2021; Wu et al., 2021), adaptive prompting (Lee et al., 2023), shared-specific modeling (Wang et al., 2023), dynamic expert mixtures (Li et al., 2025), or constraint-based robustness (Zhou et al., 2020). More recently, federated multimodal learning (MFL) has emerged to address privacy concerns in distributed settings, with methods incorporating foundation models (Che et al., 2024), meta-learning (Singh et al., 2023), open-vocabulary prototyping (Zeng et al., 2024), cross-modal prototypes (Le et al., 2025), prototype masks with contrast (Bao et al., 2023), reconfigurable representations (Nguyen et al., 2025), flexible mixture-of-experts (Yun et al., 2024), and quality-aware aggregation for incomplete sensing (Orzikulova et al., 2024) to mitigate incomplete data.

Despite these advances, critical gaps persist. To the best of our knowledge, no prior work has systematically evaluated multimodal methods under such severe missing regimes (over 50% aggregate modality absence with highly heterogeneous patterns) while maintaining strong performance across both multi-label classification and regression tasks. Most approaches remain task-specific, struggle with extreme incompleteness, or fail to seamlessly extend to federated environments where data is non-IID and only a fraction of clients participate per round. Moreover, existing evaluation metrics often fail to fairly quantify degradation under varying missing patterns. Although recent works such as Flex-MoE (Yun et al., 2024), PEPSY (Nguyen et al.) (Nguyen et al., 2025), and Orzikulova et al. (Orzikulova et al., 2024) have made important strides in handling heterogeneous or severe missing patterns, they primarily rely on indirect mechanisms such as expert routing, client-side reconfiguration signals, or quality-aware aggregation rather than explicit feature-level reconstruction.

While simulation of absent modalities from available ones is a known technique, the proposed FArms framework (with its centralized variant wArms) differentiates itself through **learnable simulating arms** that explicitly reconstruct missing modality embeddings on-the-fly. These arms consist of bidirectional pairwise simulators and multi-head-attention-based fusion simulators, supervised by direct ℓ_2 reconstruc-

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tion losses applied in a case-aware manner across all availability regimes (single, partial, and full modalities), together with an optional two-stage optimization that first stabilizes the simulators before task-specific fine-tuning. This design provides stronger instance-specific cross-modal transfer, supports both classification and regression with minimal adaptation, and naturally extends to federated settings. We include the centralized wArms variant to clearly demonstrate that the core simulating-arms mechanism delivers robust gains independently of federated challenges, while the full FArms shows how it scales under non-IID data and partial client participation.

To address these limitations, we propose FArms, a novel federated multimodal framework (with centralized variant wArm) that handles arbitrary missing modalities via learnable simulating arms, frozen pre-trained encoders, pairwise and fusion-based simulators, and a two-stage optimization strategy. Our main contributions are:

- We introduce the first framework in the literature demonstrating robust performance under extremely severe missing modalities with $>50\%$ aggregate absence.
- We show task-agnostic applicability to both multi-label multimodal classification and regression tasks with minimal adaptation.
- We perform a comparative study against strong baselines (e.g., PmcmFL (Bao et al., 2023), ShaSpec (Wang et al., 2023), M^3Care (Zhang et al., 2022), MissModal (Lin & Hu, 2023b), Flex-MoE (Yun et al., 2024), and PEPSY (Nguyen et al., 2025). (Orzikulova et al., 2024)) in heavy missing regimes and in the federated setting.
- We derive a seamless extension to federated settings, preserving or exceeding centralized quality under non-IID data and partial client participation (e.g., only half of clients commit per round) on 10 clients with just 10 rounds or fewer.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 introduces the proposed learning framework FArms. Section 4 describes datasets and the related experimental setup. Section 5 reports results and analysis, and section 6 presents a conclusion.

2. Related Work

Multi-modal learning with missing modalities has been extensively explored in centralized settings. Early works assess Transformer robustness to incomplete inputs and introduce adaptive fusion strategies (Ma et al., 2022) or missing-modality-aware prompting (Lee et al., 2023). Generation-based approaches reconstruct missing features using missing modality imagination (Zhao et al., 2021), modality translation (Liu et al., 2024), shared-private frameworks (Wu et al., 2021), duplex diffusion for unbiased recovery (Dai et al., 2025), or decoding-oriented prototype decomposition (Jin et al., 2023). Recent methods further advance representation learning through text-enhanced features (Chen et al., 2025), hybrid cross-modal interactions (Fu et al., 2024), self-supervised multi-task learning for modality-specific representations (Yu et al., 2021), cross-modal representation interactive learning (Huang et al., 2023), dynamic shifting of representations via hybrid attention (Lin & Hu, 2023a), simple dynamic mixtures of modality experts (Li et al., 2025), constraint-based robustness improvements (Lin & Hu, 2023b)(Zhou et al., 2020), and cross-modality attention adaptations for unified image-text processing under varying availability (Mirza et al., 2023).

Cross-modality consistency modeling (You et al., 2016; Liu et al., 2022) and shared-specific feature learning (Wang et al., 2023) effectively mitigate representation gaps in heavy missing regimes. Bayesian meta-learning approaches (Ma et al., 2021) enable flexible and efficient handling of severely missing modalities across training and testing.

Federated multi-modal learning (MFL) extends these ideas to privacy-preserving distributed environments. Methods address incomplete modalities via foundation model integration (Che et al., 2024), meta-learning (Singh et al., 2023), open-vocabulary multi-modal prototyping (Zeng et al., 2024), cross-modal prototypes (Le et al., 2025), or prototype masks with contrastive alignment (Bao et al., 2023). Recent advances in prominent venues introduce reconfigurable representations tailored for heterogeneous missing patterns in federated settings (Nguyen et al., 2025), cross-modal augmentation by retrieval under missing modalities (Poudel et al., 2024), feature imputation networks designed for federated reconstruction (Poudel et al., 2025), flexible mixture-of-experts for arbitrary modality combinations (Yun et al., 2024), and modality-invariant representation learning with quality-aware aggregation for time-series healthcare sensing (Orzikulova et al., 2024).

While these approaches have shown promising results, many rely on indirect alignment mechanisms (e.g., contrastive losses, expert routing in Flex-MoE (Yun et al., 2024), or reconfigurable client-side embeddings (Nguyen et al., 2025)), global or within-modality prototype matching (e.g., MFCPL (Le et al., 2024), PMCH (Zhu et al., 2025)), or quality-aware aggregation and distillation (e.g., Orzikulova et al. (Orzikulova et al., 2024)). Such strategies can be effective for specific patterns but often lack explicit per-sample reconstruction supervision across all availability regimes and may remain sensitive to extreme missing rates or task shifts.

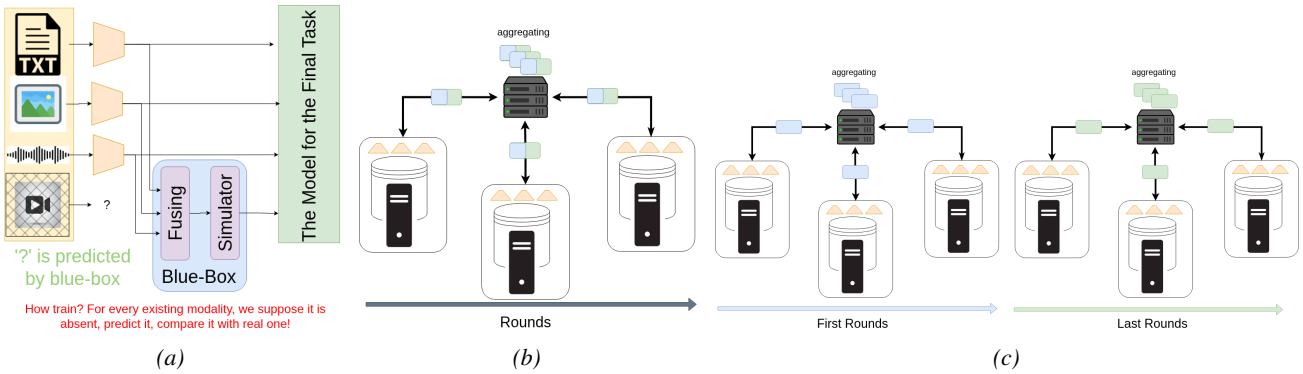

Figure 1. a) In over proposal handling missing modalities completely separated from final task. Blue-box is for imputing missing-modalities and green box is devoted to the final task, (b) In federated setting, both blue-box and green box can be optimized simultaneously or (c) blu-box can be trained in the first initial rounds, then the green box can be trained in the remaining rounds.

3. The Proposed Model

Consider a task for a K-modal dataset with modalities $\mathcal{M} = \{\mathbf{M}_1, \mathbf{M}_2, \ldots, \mathbf{M}_K\}$. Each sample consists of partial observations $(\mathbf{m}_1, \ldots, \mathbf{m}_K, \mathbf{t})$, where $\mathbf{m}_k \in \mathcal{X}_k$ or is missing, and $\mathbf{t} \in \mathcal{T}$ is the target value. The target is a class label for classification or a continuous value for regression problems. Let $\mathcal{A} \subseteq \{1, \ldots, K\}$ denote the available modalities for a sample. The goal is to learn a predictor $f : \mathcal{X}_1 \times \cdots \times \mathcal{X}_K \to \mathcal{T}$ that performs well using only the modalities in $\mathcal{A}$. The predictor reconstructs missing features during training and inference.

To handle this challenge, we propose FArms and wArms. FArms stands for **F**ederated multi-modal learning **w**ith learnable simulating **arms** for missing modalities, and wArms stands for its centralized variant (multi-modal learning **w**ith learnable simulating **arms**). The term 'arms' metaphorically captures the framework's ability to reach out and reconstruct missing modalities through multiple learnable pathways—each simulator acts as an extensible arm that grasps information from available modalities. The overall view of our proposal can be viewed in Figure 1.

3.1. The Necessary Components

For each modality k, a pretrained encoder $E_k : \mathcal{X}_k \to \mathbb{R}^{d_k}$, with frozen parameters, extracts embeddings:

$$\mathbf{M}'_k = E_k(\mathbf{M}_k), \quad \mathbf{M}'_k \in \mathbb{R}^{d_k}.$$

Frozen encoders leverage transfer learning for efficiency while keeping computational overhead low. Unavailable modalities do not produce $\mathbf{M}'_k$, which should be simulated later.

Missing modality features are simulated via trainable modules that map from available modalities to the target space. This simulation promotes modality invariance by learning shared representations in order to reduce performance degradation under missing data. For each pair of modalities (i, j) where $i \neq j$, we define bidirectional simulators:

$$\hat{\mathbf{M}}'_i = S_{j \to i}(\mathbf{M}'_j)$$

$$\hat{\mathbf{M}}'_j = S_{i \to j}(\mathbf{M}'_i)$$

where $S_{j \to i} : \mathbb{R}^{d_j} \to \mathbb{R}^{d_i}$ and $S_{i \to j} : \mathbb{R}^{d_i} \to \mathbb{R}^{d_j}$ are trainable neural networks that project features from one modality space to another. These pairwise simulators enable the model to learn direct mappings between any two modalities. When multiple source modalities are available, we leverage richer cross-modal information through fusion-based simulation. For each target modality j and a set of source modalities $\mathcal{S} = \{s_1, s_2, \ldots, s_p\}$ where $j \notin \mathcal{S}$, we first fuse the source modalities using the multi-head attention fusion mechanism:

$$\mathbf{F}_{\mathcal{S}} = f_{\mathcal{S}}(\mathbf{M}'_{s_1}, \mathbf{M}'_{s_2}, \ldots, \mathbf{M}'_{s_p})$$

where $f_{\mathcal{S}} : \mathbb{R}^{d_{s_1}} \times \cdots \times \mathbb{R}^{d_{s_p}} \to \mathbb{R}^{d_f}$ is the fusion module described in Section 3.3. The fused representation $\mathbf{F}_{\mathcal{S}} \in \mathbb{R}^{d_f}$ is then used to simulate the target modality:

$$\hat{\mathbf{M}}'_j = S_{\mathcal{S} \to j}(\mathbf{F}_{\mathcal{S}}) \tag{1}$$

3.2. The Training and Inference

The training procedure involves standard learning (e.g., minimising the classification loss) and simulation in order to learn modality-agnostic representations confidently. For each sample, we consider existing features $\{\mathbf{M}'_k\}_{k \in \mathcal{A}}$, and simulate the missing ones to form a complete set $\{\tilde{\mathbf{M}}'_k\}_k$ (existing and simulated features). These latter are fused using f_{final} which is a dedicated multi-head self-attention (MHSA) for all K modalities to predict $\hat{\mathbf{t}} = g(\mathbf{H})$ where $\mathbf{H} = f_{\text{final}}(\{\tilde{\mathbf{M}}'_k\})$, and

$$\tilde{\mathbf{M}}'_k = \begin{cases} \mathbf{M}'_k & \text{if modality } k \text{ is available} \\ \hat{\mathbf{M}}'_k \text{ (see Eq.1)} & \text{otherwise} \end{cases} \tag{2}$$

The total loss balances reconstruction fidelity and task accuracy:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{sim}} + \beta \mathcal{L}_{\text{task}}, \tag{3}$$

with $\alpha, \beta > 0$ as hyper-parameters.

Let mention that only simulators, fusion operations, and g are updated, and encoders (E_k) are kept frozen.

During training, we employ a comprehensive simulation strategy that depends on the available modality configuration for each sample. Let $\mathcal{A} \subseteq \{1, 2, \ldots, K\}$ denote the set of available modalities for a given training sample. Depending on $\mathcal{A}$, we consider the following scenarios in terms of missing modality generation:

Case 0 - Single Modality ($|\mathcal{A}| = 1$**, say** $\mathcal{A} = \{i\}$): All other modalities are learned as follows: $\hat{\mathbf{M}}'_j = S_{i\to j}(\mathbf{M}'_i)$ for $j \neq i$, yielding $\tilde{\mathbf{M}}'_j = \hat{\mathbf{M}}'_j$. No reconstruction is possible, and since no ground-truth features for $j \neq i$, so: $\mathcal{L}_{\text{sim}} = 0$. This focuses training on task performance from uni-modal inputs, with simulations refined indirectly via $\mathcal{L}_{\text{task}}$ in Eq. 3 across the dataset.

Case 1 - Subset of Modalities Available ($1 < |\mathcal{A}| < K$): When a proper subset of modalities is available, we perform two types of simulation. First, for each pair of available modalities (i, j) where $i, j \in \mathcal{A}$ and $i \neq j$, we perform bidirectional pairwise simulation and compute the reconstruction losses:

$$\mathcal{L}_{\text{sim}}^{(i,j)} = \|\mathbf{M}'_i - S_{j\to i}(\mathbf{M}'_j)\|^2 + \|\mathbf{M}'_j - S_{i\to j}(\mathbf{M}'_i)\|^2 \tag{4}$$

Second, for each missing modality $k \in \{1, \ldots, K\} \setminus \mathcal{A}$, we simulate it using fusion of all available modalities:

$$\hat{\mathbf{M}}'_k = S_{\mathcal{A}\to k}(f_{\mathcal{A}}(\{\mathbf{M}'_i\}_{i\in\mathcal{A}})).$$

This simulated feature set is then used for downstream classification, but no reconstruction loss is computed since the ground truth feature set is unavailable. The total simulation loss is the average over all pairs:

$$\mathcal{L}_{\text{sim}} = \frac{1}{\binom{|\mathcal{A}|}{2}} \sum_{(i,j)\in\mathcal{A}, i\neq j} \mathcal{L}_{\text{sim}}^{(i,j)}. \tag{5}$$

Averaging prevents dominance by larger $\mathcal{A}$. This case bridges uni-modal and full supervision to allow learning partial cross-modal dependencies.

Case 2 - All Modalities Available ($|\mathcal{A}| = K$): When all modalities are present, we perform a comprehensive simulation to strengthen cross-modal alignment. For each modality $j \in \{1, \ldots, K\}$, we simulate it from all other modalities and compute the reconstruction loss as follows:

$$\mathcal{L}_{\text{sim}}^{(j)} = \|\mathbf{M}'_j - S_{\mathcal{A}\setminus\{j\}\to j}(f_{\mathcal{A}\setminus\{j\}}(\{\mathbf{M}'_i\}_{i\in\mathcal{A}, i\neq j}))\|^2. \tag{6}$$

In addition, we perform all pairwise simulations as in Case 1 to maintain fine-grained cross-modal mappings:

$$\mathcal{L}_{\text{sim}}^{\text{pair}} = \sum_{i=1}^{K} \sum_{j=1, j\neq i}^{K} \|\mathbf{M}'_j - S_{i\to j}(\mathbf{M}'_i)\|^2. \tag{7}$$

The total simulation loss reads as follows:

$$\mathcal{L}_{\text{sim}} = \frac{1}{K} \sum_{j=1}^{K} \mathcal{L}_{\text{sim}}^{(j)} + \frac{\mathcal{L}_{\text{sim}}^{\text{pair}}}{K(K-1)}. \tag{8}$$

This comprehensive supervision ensures strong baselines for sparser cases, as full alignments propagate via shared parameters. The procedure is summarized in Algorithm 1. It processes mixed-availability batches for the sake of robustness.

Algorithm 1 General Training Procedure (Joint Optimization)

Require: Dataset $\mathcal{D}$, α, β, η
for each epoch **do**
  for each batch in $\mathcal{D}$ **do**
    Extract $\{\mathbf{M}'_k\}_{k\in\mathcal{A}}$ for each sample
    Determine case from $|\mathcal{A}|$;
    if Case 0 **then**
      Pairwise sims from single; $\mathcal{L}_{\text{sim}} = 0$
    else if Case 1 **then**
      Pairwise losses over $\mathcal{A}$: Eq.5
    else if Case 2 **then**
      Leave-one-out fusion + all pairwise losses: Eq.8
    end if
    $\{\tilde{\mathbf{M}}'_k\}$ = existing/sim for all k; $\mathbf{H} = f_{\text{final}}(\{\tilde{\mathbf{M}}'_k\})$
    $\hat{\mathbf{y}} = g(\mathbf{H})$; $\mathcal{L}_{\text{task}}$ customized and computed based on task type
    $\mathcal{L} = \alpha\mathcal{L}_{\text{sim}} + \beta\mathcal{L}_{\text{task}}$
    Update simulators/fusions/g via η using an optimizer(e.g., Adam)
  end for
end for

It is possible to perform a two-stage optimization. In the first stage, we optimize only the simulation parameters using primarily $\mathcal{L}_{\text{sim}}$ (and optionally a small weight on $\mathcal{L}_{\text{task}}$) for a number of epochs E_{sim}. After that, we freeze the simulation parameters and optimize only the task-related parameters (fusion and head) using $\mathcal{L}_{\text{task}}$. This two-stage approach can help stabilize the simulations before fine-tuning the downstream task. The procedure is summarized in Algorithm 2.

Inference procedure is straightforward. For input $\mathcal{A}$, it simulates each missing modality $k \notin \mathcal{A}$ using the fusion of all available modalities: $\hat{\mathbf{M}}'_k = S_{\mathcal{A}\to k}(f_{\mathcal{A}}(\{\mathbf{M}'_i\}_{i\in\mathcal{A}}))$, and complete set $\{\tilde{\mathbf{M}}'_k\}_k$ is obtained and fused consequently via f_{final}, and finally the prediction is made via g.

Algorithm 2 Two-Stage Training Procedure

Require: Dataset $\mathcal{D}$, η, E_{sim}
for epoch = 1 to E_{sim} **do**
  for each batch in $\mathcal{D}$ **do**
    Extract $\{\mathbf{M}'_k\}_{k\in\mathcal{A}}$ for each sample
    Determine case from $|\mathcal{A}|$
    Compute $\mathcal{L}_{\text{sim}}$ as in Algorithm 1
    Update only simulating related weights via $\mathcal{L}_{\text{sim}}$
  end for
end for
Freeze simulation parameters
for epoch = $E_{\text{sim}} + 1$ to total epochs **do**
  for each batch in $\mathcal{D}$ **do**
    Use frozen simulators to get $\{\tilde{\mathbf{M}}'_k\}$
    $\mathbf{H} = f_{\text{final}}(\{\tilde{\mathbf{M}}'_k\})$
    $\hat{\mathbf{y}} = g(\mathbf{H})$; compute $\mathcal{L}_{\text{task}}$
    Update task-related fusions and g via η (e.g., Adam)
  end for
end for

3.3. Details about the Trainable Components and Loss Functions

Each simulator $S_{\mathcal{S}\to j} : \mathbb{R}^{d_f} \to \mathbb{R}^{d_j}$ follows an multi-layer MLP architecture. A fusion-based simulator can capture more complex cross-modal relationships by leveraging complementary information from multiple source modalities. The fusion module $f_{\mathcal{S}} : \prod_{i\in\mathcal{S}} \mathbb{R}^{d_i} \to \mathbb{R}^{d_f}$ uses a multi-head self-attention or MHSA for cross-modal interactions. Inputs are projected onto d_f:

$$\mathbf{m}_i^{\text{proj}} = W_i^{\text{proj}}\mathbf{M}'_i + \mathbf{b}_i^{\text{proj}}, \quad i \in \mathcal{S}. \tag{9}$$

Stacking into sequence $\mathbf{M} = [\mathbf{m}_1^{\text{proj}}; \ldots; \mathbf{m}_{|\mathcal{S}|}^{\text{proj}}] \in \mathbb{R}^{|\mathcal{S}|\times d_f}$ and applying L MHSA layers will yield:

$$\text{head}_h = \text{softmax}\left(\frac{(\mathbf{M}W_h^Q)(\mathbf{M}W_h^K)^\top}{\sqrt{d_f/H}}\right)(\mathbf{M}W_h^V), \tag{10}$$

$$\text{MHSA}(\mathbf{M}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_H)W^O, \tag{11}$$

with residual and norm: $\mathbf{M}^{(\ell)} = \text{LayerNorm}(\mathbf{M}^{(\ell-1)} + \text{MHSA}(\mathbf{M}^{(\ell-1)}))$. Finally we compute mean-pool: $\mathbf{F}_{\mathcal{S}} = \frac{1}{|\mathcal{S}|}\sum_i \mathbf{M}_i^{(L)}$, and consequently $\text{ReLU}(W^{\text{out}}\mathbf{F}_{\mathcal{S}} + \mathbf{b}^{\text{out}})$.

This attention-based fusion dynamically weights modalities and enhances robustness to noisy or incomplete inputs. The task head $g : \mathbb{R}^{d_f} \to \hat{\mathbf{y}}$ is an N_g-layer MLP with ReLUs and dropout. For classification with $|C|$ discrete classes, $\hat{\mathbf{y}}$ are logits and

$$\mathcal{L}_{\text{task}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C}\mathbf{1}_{y_i=c}\log\left(\frac{\exp(\hat{\mathbf{y}}_{i,c})}{\sum_{c'=1}^{C}\exp(\hat{\mathbf{y}}_{i,c'})}\right) \tag{12}$$

for cross-entropy. For regression with $\mathbf{y} \in \mathbb{R}^d$, $\hat{\mathbf{y}} \in \mathbb{R}^d$ and $\mathcal{L}_{\text{task}} = \frac{1}{Nd}\sum_{i=1}^{N}\|\mathbf{y}_i - \hat{\mathbf{y}}_i\|_2^2$ for MSE. This flexibility supports diverse tasks, with $\mathcal{L}_{\text{task}}$ driving end-to-end optimization.

3.4. Extension to Federated Learning

The proposed FArms framework naturally extends to federated learning settings, such as FedAvg, where multiple clients collaboratively train a shared model without exchanging raw data. This is particularly valuable for multi-modal tasks in privacy-sensitive domains, such as healthcare or mobile sensing. Clients hold heterogeneous local datasets with varying modality availability, and centralizing data is infeasible.

Indeed, in our experiments, FArms supports a two-stage federated optimization, based on Algorithm 2. In the initial R_{sim} rounds, clients and the server focus on optimizing simulation parameters using $\mathcal{L}_{\text{sim}}$ (with optional small $\mathcal{L}_{\text{task}}$). After aggregation in these rounds, simulation parameters are frozen globally. The remaining $R - R_{\text{sim}}$ rounds optimize only task-related parameters (fusions and head) using $\mathcal{L}_{\text{task}}$. This stabilizes the cross-modal simulations before task fine-tuning, enhancing robustness in heterogeneous FL environments.

In a FedAvg setup with N clients and R communication rounds, the server initializes a global model θ^0. The trainable parameters are the simulators S, the fusion operations f, and the head g. The frozen encoders, E_k, are pre-loaded locally.

For each round $r = 1 \ldots R$, we apply the following:

- **Client Selection and Local Training**: The server selects a subset $\mathcal{C}_r \subseteq \{1, \ldots, M\}$ of clients, such as $|\mathcal{C}_r| = [10\%M]$. Each selected client $c \in \mathcal{C}_r$ receives the current global model θ^{r-1} and performs P local epochs using the training procedure in Algorithm 2 on its private dataset $\mathcal{D}_c$. Each client follows the first global loop in Algorithm 2 for $r \le R_{\text{sim}}$, which updates only the components belonging to the simulating arms, and for $r > R_{\text{sim}}$ follows the second global loop, which updates the components belonging to the task. Frozen encoders E_k remain unchanged, ensuring no modality-specific data leakage. The client computes a local update $\Delta_c = \theta_c^P - \theta^{r-1}$, where θ_c^P are the updated parameters after local local epochs P.
- **Upload and Aggregation**: Clients upload only Δ_c or θ_c^E to the server, not raw data or features. The server aggregates via weighted averaging by client dataset size $n_c = |\mathcal{D}_c|$:

$$\theta^r = \sum_{c\in\mathcal{C}_r}\frac{n_c}{\sum_{c'\in\mathcal{C}_r} n_{c'}}\theta_c^E. \tag{13}$$

This yields the new global model θ^r.

- **Broadcast**: The server broadcasts θ^r back to all clients for the next round.

This extension preserves data privacy, as only parameters are exchanged, and handles FL-specific challenges. Heterogeneity arises because local $\mathcal{D}_c$ varies; for example, in a multi-modal mobile FL, mobile clients may lack audio. Simulations ensure consistent K-modal fusion per client. Efficiency is maintained since frozen encoders reduce local compute and simulations are lightweight MLPs. Convergence is supported by the dual loss $\mathcal{L}$, which stabilizes the updates and mitigates non-IID issues from missing modalities. After R rounds, the final θ^R is deployed globally. Empirical extensions, such as to FedProx for straggler handling, are straightforward, as the core loop is modular.

3.5. Theoretical Communication and Computational Cost Analysis

While any-to-any modality simulation can increase overhead, FArms keeps costs low through frozen pretrained encoders and lightweight trainable components (MLPs for simulators and compact MHSA for fusion). The optional two-stage optimization further reduces communication: only simulator/fusion parameters are updated and aggregated in the first R_{sim} rounds; after freezing simulators, only the small task head and final fusion are communicated.

Table 1 summarizes the theoretical communication cost per round (upload + download) and key characteristics for FArms versus representative baselines. Let P denote the total trainable parameters of a baseline, d_f the fused embedding dimension, and assume K modalities with average simulator/fusion size much smaller than full encoders.

Here, $\alpha < 1$ reflects the reduced parameter set in FArms (especially after freezing simulators). Because simulators are simple MLPs ($O(d_f d_j)$ per call) and fusions use lightweight attention, both per-client computation and overall communication are lower than methods that update full modality encoders, maintain large prototype banks (PmcmFL), or route through many experts (Flex-MoE). Empirical results in Section 5 confirm that FArms achieves competitive or superior performance with reduced communication overhead, particularly under severe missing regimes.

On the client side, computational complexity per sample remains linear in the number of available modalities and embedding dimensions, avoiding the quadratic or higher costs associated with large prototype contrastive banks or complex expert routing.

4. Datasets

We evaluate our FArms approach on two widely-used multi-modal datasets that present distinct challenges: MM-IMDb for multi-label movie genre classification and CMU-MOSI for sentiment intensity regression. To assess robustness under realistic conditions, we simulate missing modality scenarios throughout both for training and testing.

MM-IMDb. The multi-modal IMDb dataset comprises image (movie posters) and text (plot summaries) modalities for movie genre classification across 27 distinct genres including Drama, Comedy, Thriller, and Romance. Each sample can be associated with multiple genres simultaneously, making this a challenging multi-label classification task. To simulate missing modalities, we configure datasets with various availability patterns:

- α-*image*-β-*text*: α% image with β% text, and one of the α or β should be 100, or complex setting should be considered,
- *complex*-γ-α -β: γ% both-modalities-presence, α% image only and β% text only.

We apply CLIP encoders to extract representations from both modalities, using standard ImageNet normalization for 224×224 images.

CMU-MOSI. The CMU multi-modal Opinion Sentiment Intensity dataset contains three modalities: text (transcripts), audio (waveforms), and video (facial expressions) from YouTube movie review segments. Each segment is annotated with a continuous sentiment score ranging from -3 (strongly negative) to +3 (strongly positive), forming a regression task. Missing modality configurations specify availability percentages for each modality independently or complex distributions defining proportions for all-three-present, single-modality, and pairwise-modality combinations:

- *X-text-Y-audio-Z-video*: *X*% text, *Y*% audio, and *Z*% video,
- *complex-TAV-T-A-V-TA-TV-AV*: *TAV*% all-modalities-presence, *T*% text only, *A*% audio only, *V*% video only, *TA*% text with audio but no video, *TV*% text with video but no audio, *AV*% audio with video but no text.

5. Results

We report FArms results on MM-IMDb and CMU-MOSI. FArms uses FLOWER (Beutel et al., 2020) for federated learning. Data is heterogeneously distributed over 10 clients, with only 50% participating per round. More details about parameters for wArm and FArms would be described in each experiment.

As shown in Table 2, FArms significantly outperforms all baselines on both datasets across nearly all missing modality

Table 1. Comparison of communication and computational characteristics.

Method	Communication per Round	Trainable Components	Key Cost Factor
FArms (Ours)	$O(N \cdot \alpha P_{\text{FArms}})$	Lightweight MLPs + small MHSA	Two-stage (simulators frozen after R_{sim})
PEPSY (Nguyen et al., 2025)	$O(N \cdot P_{\text{PEPSY}})$	Client-side embedding controls + reconfigurable modules	Per-client missing-pattern embeddings
Flex-MoE (Yun et al., 2024)	$O(N \cdot P_{\text{FlexMoE}})$	Mixture-of-Experts + missing modality bank	Expert routing parameters
PmcmFL (Bao et al., 2023)	$O(N \cdot P_{\text{PmcmFL}})$	Prototype library + contrastive modules	Global prototype maintenance and masks

configurations. On MM-IMDb, FArms achieves remarkably stable and high F-micro scores (**0.93–0.94**) even under severe and complex missing patterns, while baselines such as ShaSpec, M^3Care, and PmcmFL show clear degradation. On CMU-MOSI, FArms also obtains the best performance in most settings, demonstrating strong robustness to heavy modality incompleteness.

Table 3 presents similar trends. FArms achieves the highest F-macro scores in almost all configurations, except the fully available case on MM-IMDb where PmcmFL obtains a slightly higher score.

Figure 2 qualitatively evaluates imputed embedding quality in a complex missing-modality setting via t-SNE projection to 2D. Yellow points mark the ground-truth embeddings from present modalities; red and blue denote the imputed image and text embeddings, respectively. Baselines ShaSpec, M^3Care, and PmcmFL produce imputed embeddings that cluster separately from the ground truth, indicating significant manifold deviation and potential distributional shift. FArms, however, yields imputed points that tightly overlap and integrate with ground truth in both modalities, preserving realistic distribution.

A Note on Communication Overhead FArms converges effectively in just 10 communication rounds, even under non-IID data and partial client participation (half of clients per round). This low round count reduces communication overhead substantially compared to typical federated multimodal methods requiring dozens to hundreds of rounds on CMU-MOSI and MM-IMDb (Le et al., 2025; Yuan et al., 2024). For example, FedMVP (Che et al., 2024) uses 200 rounds on CMU-MOSEI, and MFCPL (Le et al., 2025) uses 50 on CMU-MOSI. Therefore, it can be considered a strong point for FArms that with 10 communication rounds, it can also beat the baseline central method. The advantage is not limited just to this point, but FArms can also obtain favourable results when the communication round is 5, as

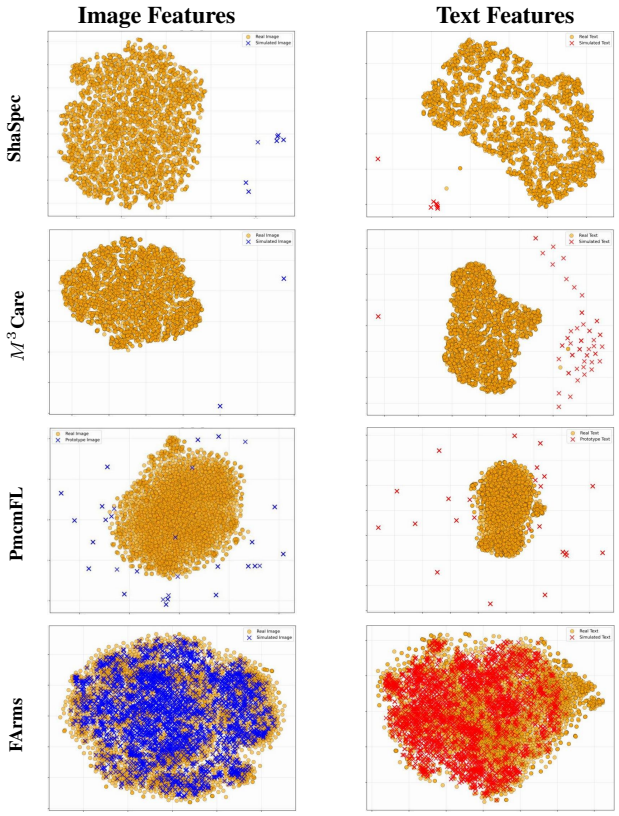

Figure 2. Visualisation of real vs simulated feature embeddings for missing modalities across different methods.

shown in Fig. 4. This figure highlights results for severely missing configurations, and for other configurations we have similar diagrams. Furthermore, since FArms employs frozen encoders, it does not need to have them in communications, as every client can employ the same encoder with the same weights as done in our experiments.

6. Conclusion

In this work, we presented FArms, a novel federated multimodal learning framework that robustly handles arbitrary

Table 2. F-micro obtained by the algorithms on MM-IMDb and CMU-MOSI

	MM-IMDb						CMU-MOSI							
Method	100-image-100-text	100-image-20-text	20-image-100-text	complex-10-45-45	complex-20-40-40	complex-30-35-35	100-text-100-audio-100-video	100-text-100-audio-20-video	100-text-20-audio-100-video	100-text-20-audio-20-video	20-text-100-audio-100-video	20-text-100-audio-20-video	20-text-20-audio-100-video	complex-20-20-20-10-10-10
ShaSpec	0.49	0.39	0.45	0.39	0.38	0.43	0.35	0.33	0.32	0.30	0.28	0.25	0.27	0.29
M^3Care	0.46	0.39	0.43	0.43	0.37	0.45	0.36	0.34	0.33	0.31	0.29	0.26	0.28	0.30
pmcmfl	0.66	0.51	0.48	0.50	0.51	0.54	0.38	0.36	0.35	0.33	0.31	0.28	0.30	0.32
Flex-MOE	0.11	0.09	0.10	0.08	0.09	0.10	0.61	0.54	0.51	0.51	0.49	0.43	0.49	0.51
PEPSY	0.13	0.11	0.12	0.10	0.11	0.12	0.74	0.72	0.71	0.62	0.64	0.55	0.55	0.55
FArm	**0.94**	**0.93**	**0.93**	**0.93**	**0.93**	**0.93**	**0.82**	**0.81**	**0.82**	**0.81**	**0.67**	**0.57**	**0.67**	**0.71**
Winner (t-test)	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms
Winner (WCOX)	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms

Table 3. F-macro obtained by the algorithms on MM-IMDb and CMU-MOSI

	MM-IMDb						CMU-MOSI							
Method	100-image-100-text	100-image-20-text	20-image-100-text	complex-10-45-45	complex-20-40-40	complex-30-35-35	100-text-100-audio-100-video	100-text-100-audio-20-video	100-text-20-audio-100-video	100-text-20-audio-20-video	20-text-100-audio-100-video	20-text-100-audio-20-video	20-text-20-audio-100-video	complex-20-20-20-10-10-10
ShaSpec	0.5	0.38	0.44	0.39	0.38	0.43	0.33	0.31	0.30	0.28	0.26	0.23	0.25	0.27
M^3Care	0.45	0.38	0.43	0.42	0.36	0.44	0.34	0.32	0.31	0.29	0.27	0.24	0.26	0.28
pmcmfl	**0.66**	0.49	0.45	0.48	0.49	0.54	0.36	0.34	0.33	0.31	0.29	0.26	0.28	0.30
Flex-MOE	0.08	0.06	0.07	0.05	0.06	0.07	0.57	0.51	0.48	0.48	0.46	0.40	0.46	0.48
PEPSY	0.09	0.07	0.08	0.06	0.07	0.08	0.70	0.68	0.67	0.59	0.61	0.52	0.52	0.52
FArm	0.64	**0.57**	**0.60**	**0.57**	**0.57**	**0.58**	**0.82**	**0.81**	**0.82**	**0.81**	**0.67**	**0.57**	**0.67**	**0.71**
Winner (t-test)	pmcmfl	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms
Winner (WCOX)	pmcmfl	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms	FArms

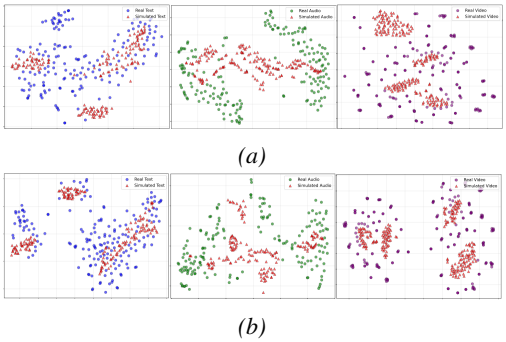

Figure 3. Visualization of real vs simulated feature embeddings for missing modalities in first run (a) and ninth run (b) for complex configuration yield by wArms

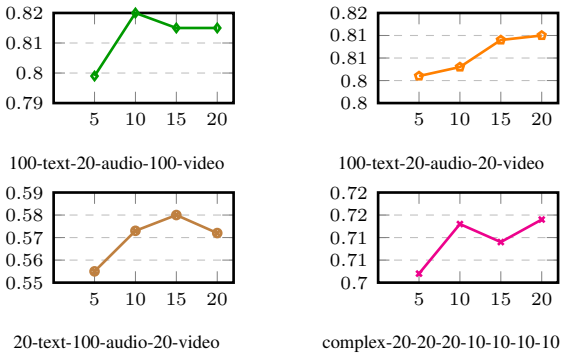

Figure 4. F-Micro changes with number of rounds across different data configurations in federated setting.

missing modalities through bidirectional simulation arms. By combining frozen pretrained encoders, pairwise and fusion-based cross-modal simulators, and a carefully designed two-stage training strategy, FArms learns modality-agnostic representations that generalize effectively across training, validation, and inference under heterogeneous availability patterns.

Extensive experiments on MM-IMDb and CMU-MOSI demonstrate the effectiveness of the approach. On MM-IMDb, FArms achieves near-perfect F-micro scores above 0.93 across all full and heavily missing configurations — with relative errors under 1% — substantially outperforming strong centralized baselines such as ShaSpec, M^3Care, and PmcmFL. The high fidelity of imputed embeddings is further confirmed by t-SNE visualizations showing tight

integration with ground-truth manifolds. On CMU-MOSI, the centralized wArms variant significantly surpasses the specialized MissModal baseline in half of the evaluated missing-modality scenarios. While the existing federated methods in literature on less than 5 clients, with dozens to hundreds of communication rounds, are still beaten by typical central methods, the fully federated FArms on 10 clients, with 10 rounds or fewer, still delivers competitive performance compared to centralized state-of-the-art methods.

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A. Supplementary - Setup

For MM-IMDb, FArms employs frozen CLIP ViT-B/32 encoders producing 512-dim embeddings, bidirectional two-layer MLP simulators $S_{1\to2}$ and $S_{2\to1}$ with hidden size 256 and ReLU, modality-specific projections to 256-dim shared space, element-wise mean fusion or single-modality pass-through, and a 3-layer MLP head $256 \to 128 \to 64 \to 27$ with ReLU and 0.1 dropout.

Federated training uses multiple rounds of 10 local epochs per selected client. We adopt the two-stage protocol from Algorithm 2: the first two rounds are used to optimize simulation parameters only, while the remaining rounds are used to update the fusion and the classification head only. Both stages employ separate Adam optimizers with learning rate 5×10^{-5}. Total federated rounds: 4.

CMU-MOSI is a common benchmark for multi-modal sentiment analysis, typically formulated as regression with scores from -3 to $+3$. Following common practice in the literature, we use the binary classification version: negative for scores from -3 to 0 and positive for scores above 0. To the best of our knowledge, no prior work simultaneously addresses federated learning, multi-modality, and missing modalities on this dataset. Among related methods, MissModal (Lin & Hu, 2023b) is specifically designed for handling missing modalities on CMU-MOSI. It is originally trained on complete data but can process missing modalities at inference time and outperforms several baselines in that setting.

For fair comparison, we include both the centralized version of our proposal, wArms, its variant trained on complete data (wArm/C), and its federated variant (FArms) against MissModal. All methods scan the training data exactly 20 times, so wArms(/C) are trained for 20 epochs, while FArms performs 10 communication rounds with 10 clients, 8 local epochs per round, and 50% client participation per round, resulting in equivalent total data exposure. We apply the following frozen encoders for wArms and FArms: BERT encoder with weights 'bert-base-uncased' for text, WavLM-Large encoder with 'microsoft/wavlm-large' weights for audio, and CLIP vision encoder with 'openai/clip-vit-base-patch32' for video frames.

B. Supplementary Results on CMU-MOSI

Table 4 reports average performance over 10 independent runs, together with statistical significance (paired t-test and Wilcoxon signed-rank test). The centralized method, wArms, is explicitly designed for multi-modal data with heavy missing-modality samples. It significantly outperforms MissModal in 4 out of 8 configurations (and matches or exceeds it in 3 additional cases) according to both statistical tests. In the federated setting, despite typical performance degradation in distributed environments, FArms still significantly outperforms MissModal in 2 out of 8 configurations. These results demonstrate that FArms maintains strong robustness to missing modalities even under realistic federated constraints. For FArms, with 10 rounds and 8 local epochs, with only 50% of clients participating per round, it scans the entire data 20 times, like other central methods with the number of epochs 20.

Table 4. F-micro comparison across missing modality settings on CMU-MOSI. Results are averaged over 10 independent runs. Statistical significance is assessed via paired t-test and Wilcoxon signed-rank test. *NO-Diff* indicates no statistically significant difference; otherwise the winning method is named.

Configuration	**MissModal**	**wArms**	**wArms/C**	**FArms**	**MissModal vs.**		
					wArms	**wArms/C**	**FArms**
100-text-100-audio-100-video	0.80	0.82	0.82	0.82	wArms	wArms/C	FArms
100-text-100-audio-20-video	0.80	0.81	0.80	0.81	NO-Diff	NO-Diff	NO-Diff
100-text-20-audio-100-video	0.80	0.81	0.81	0.82	wArms	NO-Diff	FArms
100-text-20-audio-20-video	0.81	0.81	0.80	0.81	NO-Diff	NO-Diff	NO-Diff
20-text-100-audio-100-video	0.67	0.67	0.67	0.67	NO-Diff	NO-Diff	NO-Diff
20-text-100-audio-20-video	0.57	0.68	0.64	0.57	wArms	wArms/C	NO-Diff
20-text-20-audio-100-video	0.68	0.66	0.66	0.67	NO-Diff	MissModal	NO-Diff
complex-20-20-20-10-10-10-10	0.71	0.73	0.73	0.71	wArms	wArms/C	NO-Diff